AI Training

Agility Master

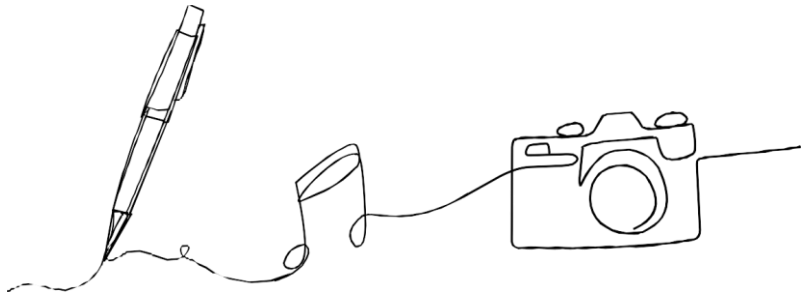
Prof. Dr. Raphael Herding

Prompt Engineering Foundations

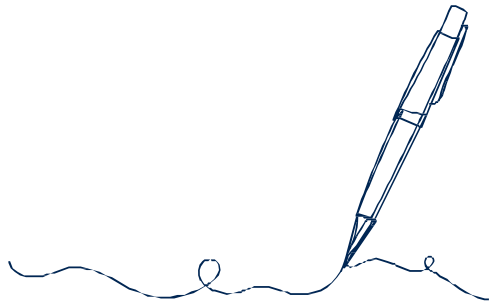
Refresher: What is Generative AI?

Emphasizing aspects relevant for Prompt Engineering

- New type of AI application
- Generates novel content on demand, using training data
- Unprecedented, adapting capabilities
- Capable of semantic & conceptual understanding
- Learns and evolves further
- Now most tools: multimodal
- Market segment with hyper-evolution

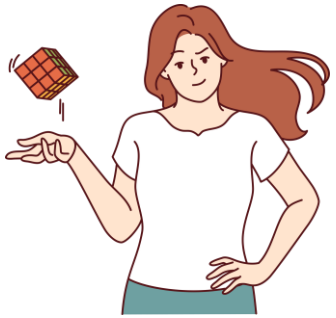


What is a Prompt?



- Input to an AI model in natural language
  - AI model understands query semantically
  - Then performs desired task

  → *keyword searches are NOT prompts*
- Via text interface, API, ...or microphone
- Sequence of prompts: *Prompt Chains!*



Benefits & Limitations



- Describe tasks in NL
  → *precisely & creatively*
- No expert skills needed
  → *e.g. Boolean; Code*
- Multi-lingual inputs

- Limited no. words per prompt
- Limited prompts per chain
- Lack of transparency
- Lack of repeatability

…many more

What is Prompt Engineering?

= Craft of guiding GenAI to produce desired outcomes, efficiently using natural language instructions.

- Foundational AI literacy skill
- Synonyms: Prompting; Prompt Design

5 Purposes:

- 1: Maximize consistency; improve reliability & repeatability
- 2: Leverage an AI model's sophisticated capabilities
- 3: Individualize GenAI outcomes → from generic to specific
- 4: Address GenAI-related pitfalls
- 5: Safety-tests; Jailbreaking

What is Prompt Engineering?

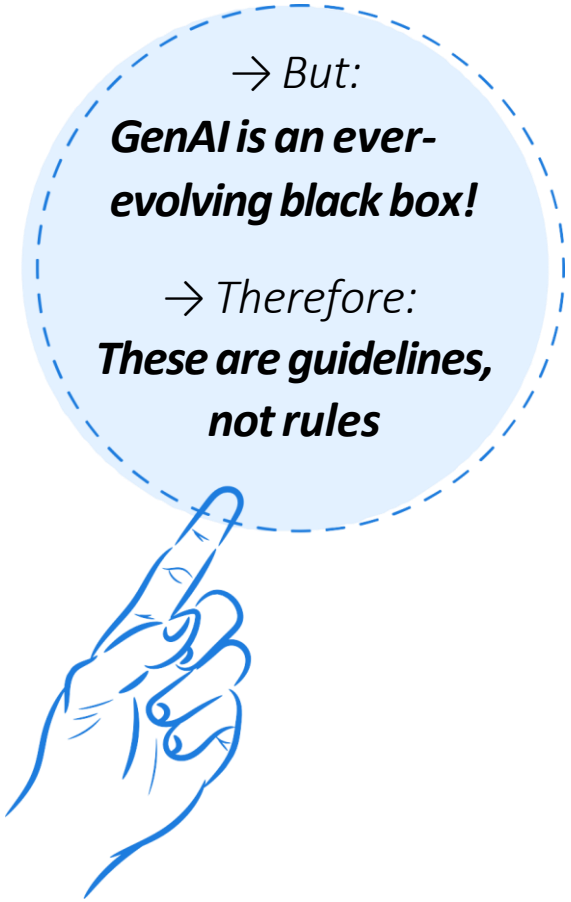
Role of Prompts in GenAI

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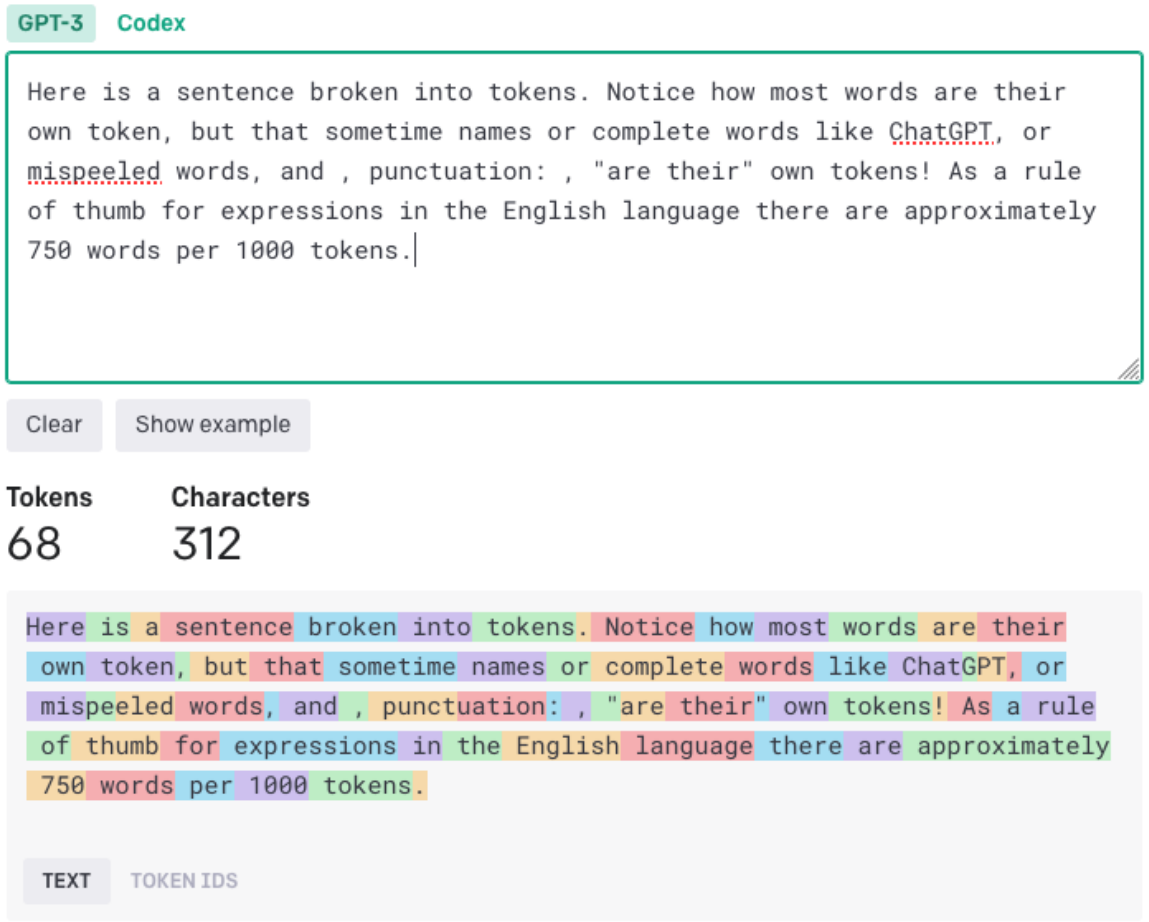
Large Language Models
When to open a new chat?

- When do we open a new chat?

Large Language Models
Token Limit

- Token limits limit the input per prompt.
- Tokens **are not words** (could reach from 1 char to entire word).

- Token limit is no real issue anymore!
  - GPT 5.4: 1.05 million





BLOCK 1

Tool Mastery: Kiro

Goal: configure Kiro so it delivers team-consistent output from the first prompt

KEY TERMS

Steering Files



Steering file

a markdown file that Kiro reads automatically before every single prompt in this repo, so you don't have to re-explain your conventions each time



Repo-level

applies to the whole project automatically, as opposed to something you'd have to repeat in each individual chat message

Steering Files



- Persistent, repo-level instructions under `.kiro/steering/` - read automatically on every prompt
- `tech.md`: language, framework, key libraries, testing conventions
- `structure.md`: folder layout, where a given kind of change belongs
- **Rule:** describe the codebase as it is, not as you wish it were - an inaccurate steering file makes Kiro confidently wrong

```
tech.md:        "Language: Python 3.12. Testing: pytest, AAA pattern,
                 mock every call in integrations/."
structure.md:   "New business rule -> services/. New endpoint -> api/,
                 calling into services/, never querying the DB directly."
```

KEY TERMS

Prompt-Files



Prompt-file

a prompt you write once, save, and then trigger again and again (like a saved macro or snippet), instead of retyping it from scratch every time

Prompt-Files



- Some prompts you write once; others you reuse dozens of times, worded consistently
- A prompt-file is a saved, reusable template invoked on demand (e.g. `/generate-unit-tests`)
- Two parts:

  Fixed part: what it always asks for.

  Variable part: what changes per invocation

- Still needs the same validation as any other AI output - a good template doesn't guarantee a good result

EXAMPLE

Fixed: "write AAA-style tests, mock every dependency in `integrations/`, match this repo's conventions"

Variable: "...for `{target_file}`"

BEFORE CHALLENGE 1

Introduction

Get in touch with Kiro

Set up your Kiro workspace, compare three ways of asking, and create your first steering files.

Techniques
Build the Context

- Ask the LLM to summarize uploaded or accessed files first.
- This builds the context so that the LLM knows the context of the current chat automatically.

- **Prompt Formula**
  - Look into the attached file and summarize what the file is about.

CHALLENGE 1

Hello, Kiro

"Kim just gave you Kiro and your team's folder. Get to know both, then teach Kiro your team."

Let Kiro summarize the team files, then write a steering file and compare the answers.

7 Principles For Good Prompt Design

1 Be concise.

- Aim for brevity & clarity
- Use language that is as simple as possible

- **Cluttered prompt & question format**

  "Can you provide me with a detailed explanation of the photosynthetical process and the significance of this biochemical reaction?"



- **More concise prompt & instructive format**

  "Explain the process of photosynthesis and its significance in detail."

2 Be clear.

- Avoid vague & ambiguous wording
- Be as specific as necessary

- **Prompt in ambigious wording and question format:**

  "How do I produce a paper?"

- **Clearer, less ambigious and imperative format:**

  "Explain the common steps of writing a research paper for a peer reviewed academic journal."



3 Include context & logical structure.

- Provide context to improve reasoning
- Build structured & coherent prompts

- **Unstructured prompt without context:**
  "Mention tasks involved in writing a research paper."



- **Structured prompt, including context:**
  "List the steps involved in writing a research paper for a student assignment. Begin with selecting a topic, end with proofreading the final draft."

4 Break down complex tasks.

In one prompt, avoid...

- combining several tasks
- asking for various formats

Divide multi-step or -topic tasks...

- into separate prompts
- or prompt chains

EXAMPLE

*"**Design** a comprehensive marketing campaign for a new eco-friendly product, detailing the target audience, key messaging, and promotional strategies. **Provide specific examples** of advertising mediums such as social media, print, and television, and create a visual mockup of an advertisement in a modern minimalist style. **Write** meaningful alt-text for the advertisement, summarizing its content for enhanced accessibility. Finally, **draft** a 2-minute pitch script for presenting this marketing campaign to potential investors."*

5 Specify desired output.

- Style & tone
- Depth & length
- Format, language, or content type
- Temperature → controls precision & creativity

- **Style & Tone:**
  "Draft a paragraph on XYZ in the simplistic style of a tabloid paper. Use a sensational tone."
- **Depth & length:**
  "Provide a high-level legal argumentation on the topic of XYZ that does not exceed 500 words."
- **Format, language, content type:**
  For example: A three-column table; a poem in Swahili; a script in Python
- **Temperature:**
  Balance creativity & precision, depending on task



6 Reflect & adapt your approach.

- Continuously evaluate outcomes
- Be flexible, fine-tune, & improve
- Adjust, then re-run!

- By applying the other principles
- By using a different prompting technique



→ *No improvement? Even with adjustments? ---* ***Start fresh!***

7 Combine these principles.

- These principles complement each other
- Mix and match modularly for optimum results



- **Earlier example, already improved (tone and style):**

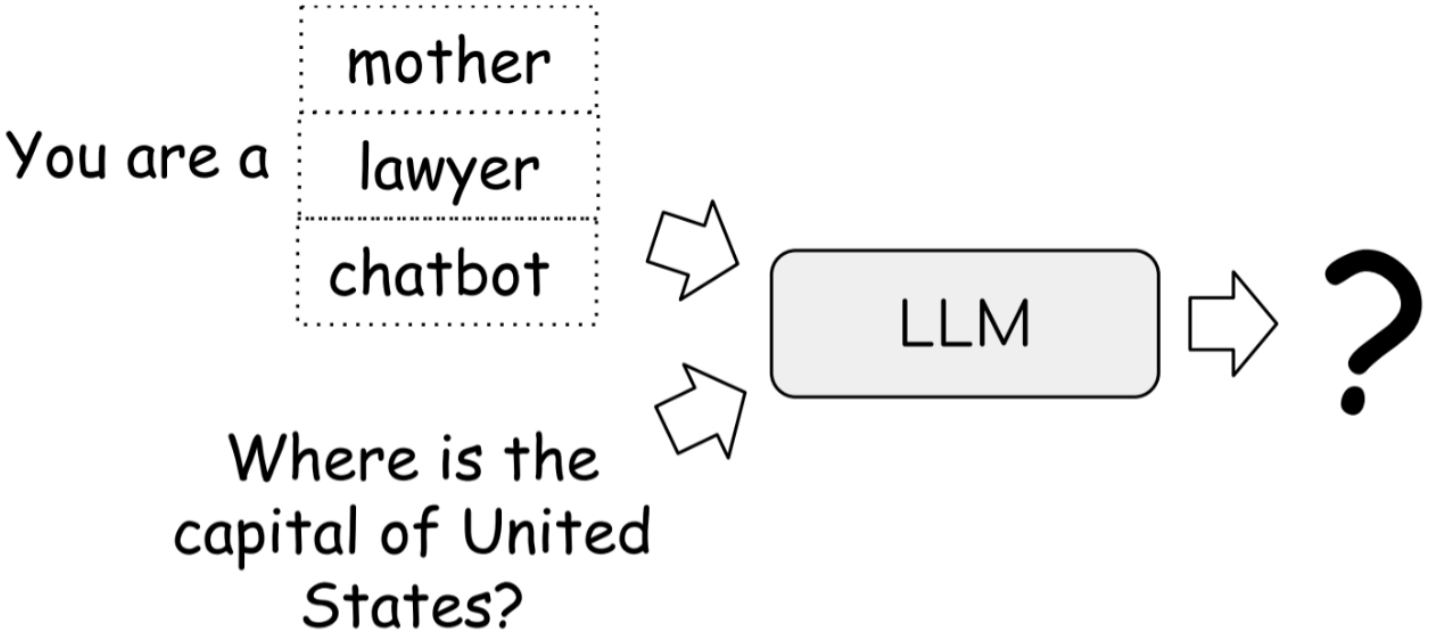
"Draft a paragraph on climate change in the simplistic style of a tabloid paper. Use a sensational tone."

- **Improved further: clarity, context, temperature, length:**

"Draft a newspaper paragraph of 500 words including a headline, on the effects of climate change. Apply the simplified grammar of a tabloid paper, and use sensational language. Be creative when describing climate change's effects."

Techniques
Role Playing/Prompting

- **Role Prompting** assigns a persona to an LLM.
- It enhances text clarity and accuracy.
- Is useful for a wide range of tasks like writing, reasoning, and dialogue-based applications.



Techniques
Role Playing/Prompting

Prompt Copy

```
You are a salesperson. Write a quick outreach email to [person] about partnering up.
```

AI Output Copy

```
Subject: Let's Discuss a Potential Partnership Opportunity
Hi [Person's Name],
I’m [Your Name] from [Your Company], and I wanted to reach out because I see a great opportunity for us to collaborate. Our products/services align well with what you’re doing at [Their Company], and I believe that by partnering, we can achieve even greater results together.
I’d love to chat about how we can create value for both of our businesses. Are you available for a quick call sometime this week?
Looking forward to your response.
```

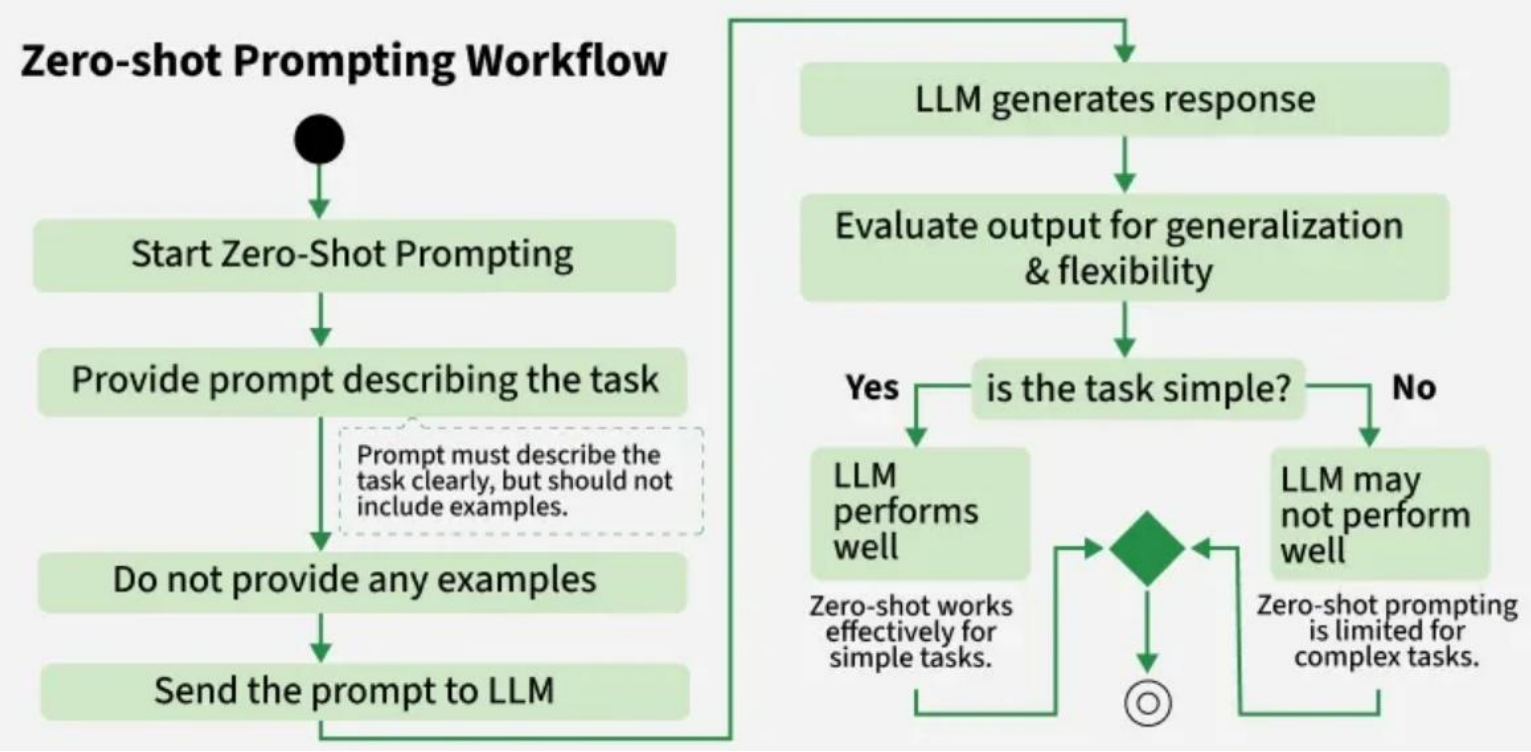
Techniques
Define the Output

- LLMs tent to output much text with less meaning.
- Specify the output as tight as possible to reduce overhead.

- **Prompt Formula**
  - Provide me information of the historic development of LLMs. Output the result in a table where in one column the date is provided and in the other the milestone of the history. Limit the milestone description to a maximum of 200 words.

Techniques
Zero-shot Prompts

- Simple prompt (instruction/question) will be executed without examples.
- Helps **quickly** creating prompts if instruction/question **is well trained**.
- If task is complex or not well trained, we have to provide examples (few-shot prompts).



Techniques
Few-shot Prompts

- **One Shot**: One piece of data or guideline given before completing request.
- **Few Shot**: Multiple pieces of data, examples or guidelines given before completing request.
- Is a technique for LLMs to generate more desired output after first viewed a few examples.

One-Shot

You give CoRecruit one example of the output you want.

[Use this example as the template:

(Paste example submission here)]

Few-Shot

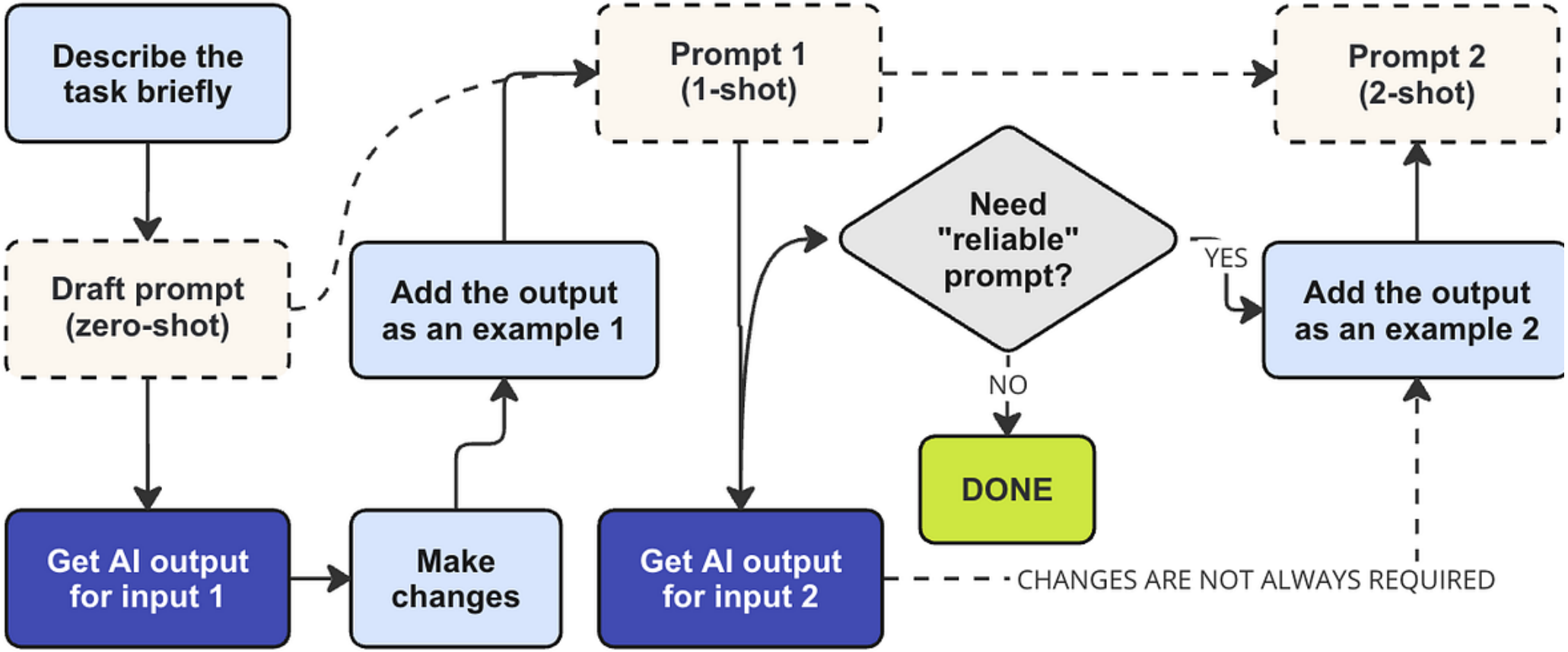
You give CoRecruit multiple examples.

[Create a candidate submission using these examples:

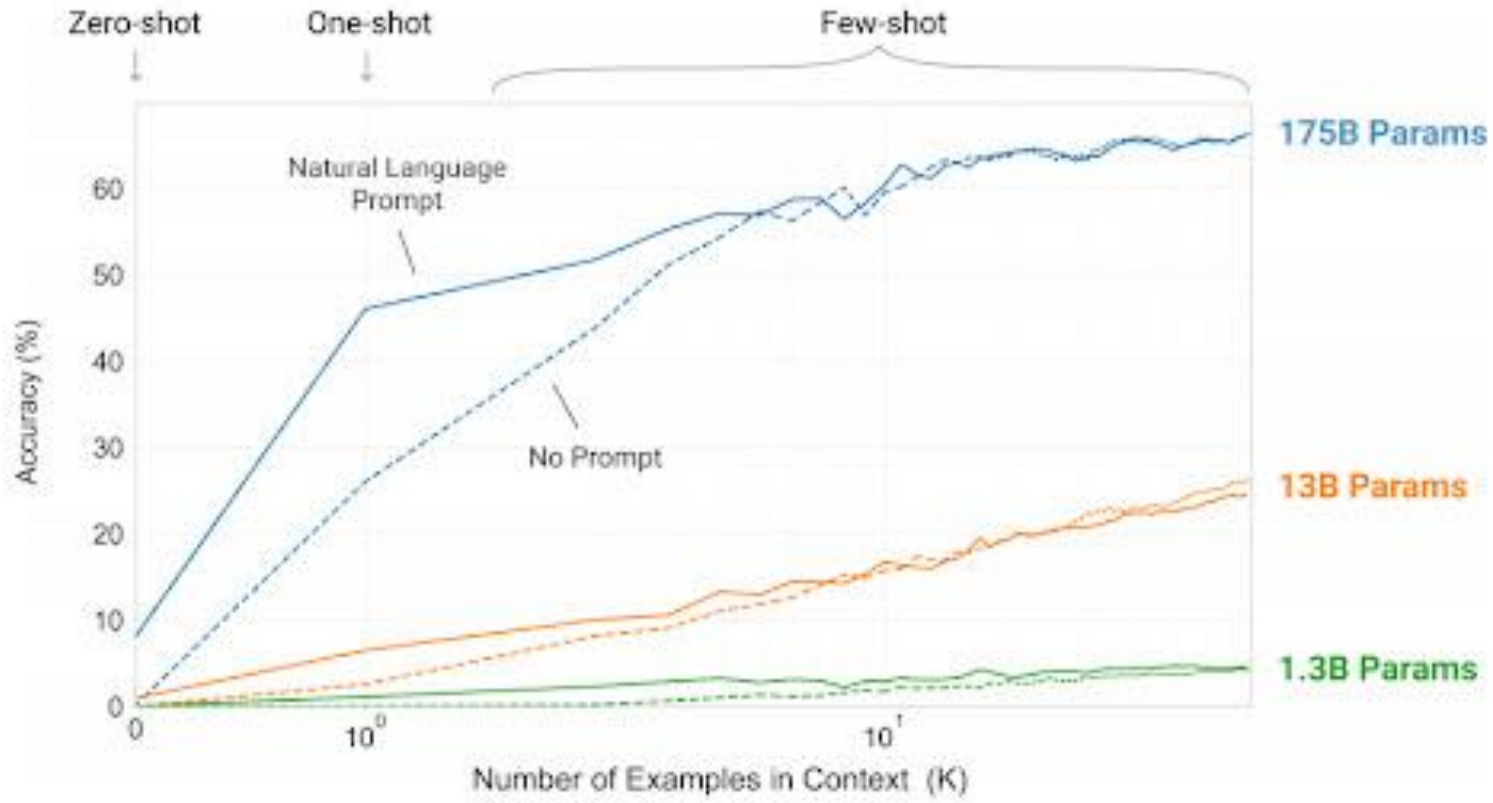
1. (Tech example)
2. (Sales example)
3. (Finance example)]

Techniques
Few-shot Prompts

- **If you don't have many examples, let the LLM create them for you**



Techniques
Context and Accuracy



- As more context you provide in the input, as better the transformer can derive the context.
- As more „shots"/prompts you provide, as more accurate the result will be.

CHALLENGE 2

The Monday Inbox

"*Five colleagues want five precise answers before 10am, one of them a risk message for the steering committee.*"

Answer all five with the 7 principles, role prompting and a defined output.

Techniques
Hallucination Guard

- Why LLMs hallucinate?
  - Limited access to real-time knowledge
  - Gaps in training data
  - Ambiguous or open-ended prompts
  - Overconfidence in generated text (LLMs don’t understand uncertainty the way humans do. They might present a wrong answer with the same confidence as a correct one)
  - … many more …

Techniques
Hallucination Guard

- They are fundamentally **probabilistic** prediction engines, not reasoning databases.
- **Next-Token Prediction**: LLMs are trained to guess the most *statistically probable* next word based on training data. They **do not "understand**" the concepts of **natural language**.

- They simply reproduce the **mathematical patterns** of how engineers or scientists typically write about those topics.
- When “no knowledge” of a topic is available, the probability distribution of a word still provides next words. Even if they are not fitting in the current context.

Techniques
Hallucination Guard

FEATURE	DESCRIPTION
Encourage abstention and “I don’t know” responses	Allow the model to say “I don’t know” and avoid speculations.
Structured, Chain-of-Thought Reasoning	Step-by-step reasoning to improve consistency in responses.
Grounding with “According To”	Use explicit references to ground responses on.
RAG with Explicit Instruction and Context	Explicitly instruct the model to rely on evidence retrieved.
Output Constraints and Limiting Scope	Restrict format and length of responses to minimize speculative elaboration and make answers more verifiable.
RAG-based Correction Loops: Chain-of-Verification	Tell the model to verify its own outputs against retrieved knowledge.
Domain-Specific Prompts, Disclaimers, and Safety Guardrails	Constrain prompts with domain rules, domain requirements, or disclaimers in high-stakes scenarios.

Techniques
Direct the LLM to Perform a Deep Analysis

- Direct the LLM to deeply analyze the result before it outputs it.
- Takes longer but the output is more precise.

- **Prompt Formula**
  - Look into the attached file and summarize what the file is about. Ensure you have read any single word of the file. Don’t miss a single word. This is a complex task so take your time and think as long as you need. Ensure you double checked the result before you output it.

Techniques
Ensure all Content is Analyzed

- Direct the LLM so that it is forced to read all pages of a document or source completely.
- ensures a much higher granularity because the LLM usually is not reading the entire document.

- **Prompt Formula**
  - Look into the attached file and summarize what the file is about. Ensure you have read any single word of the file. Don’t miss a single word.

CHALLENGE 3

The Near-Miss

"An Agility Master almost announced a SAFe rule that doesn't exist, and one Definition of Done contains a step nobody agreed on."

Check every AI claim against the sources and find the invented step.

AI Governance
Internal AI Policies

- Company rules come first: check them before you prompt
- When in doubt, ask before you paste work data

Before you prompt, check:



Approved AI tool?



Data class allowed in prompts?



Labeling rule for AI content?

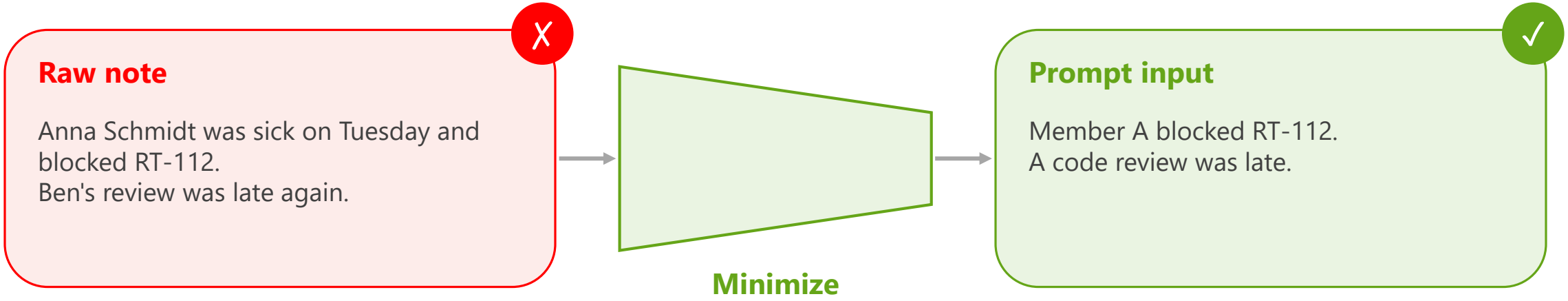


Works agreement on AI?

Clarification and Decision Support

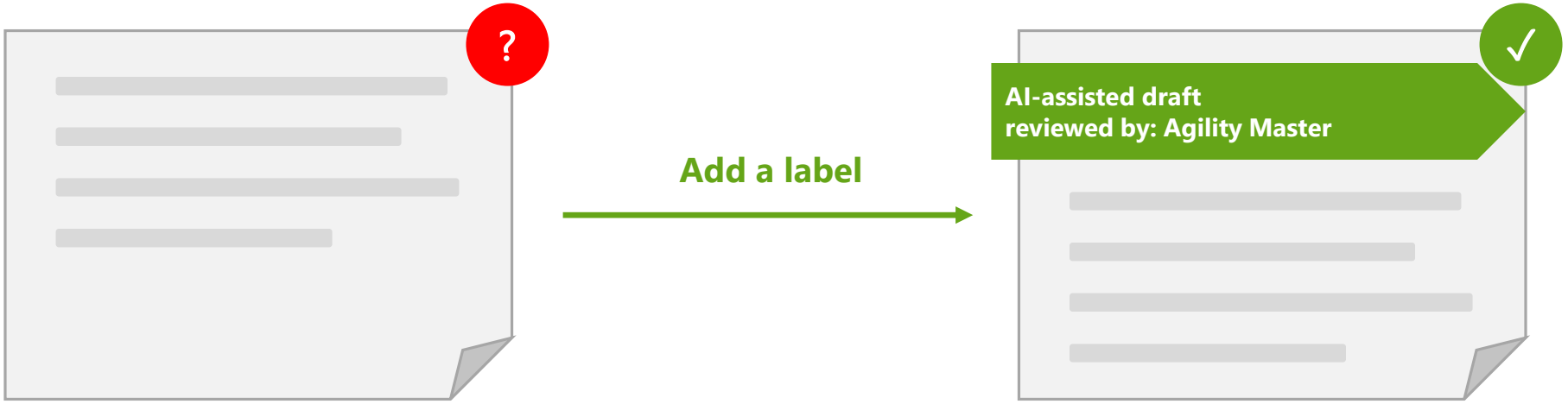
AI Governance
Data Minimization in Prompts

- Share only the data the task really needs
- Replace people with pseudonyms: Member A, B, C
- Never: health data, performance judgments, customer data, (**look into your compliance rules**)



AI Governance
Transparency and Labeling

- Say when AI helped: to the team, stakeholders and customers
- Label every AI-assisted document with its human reviewer
- Duties come from law (EU AI Act) and from companies internal rules

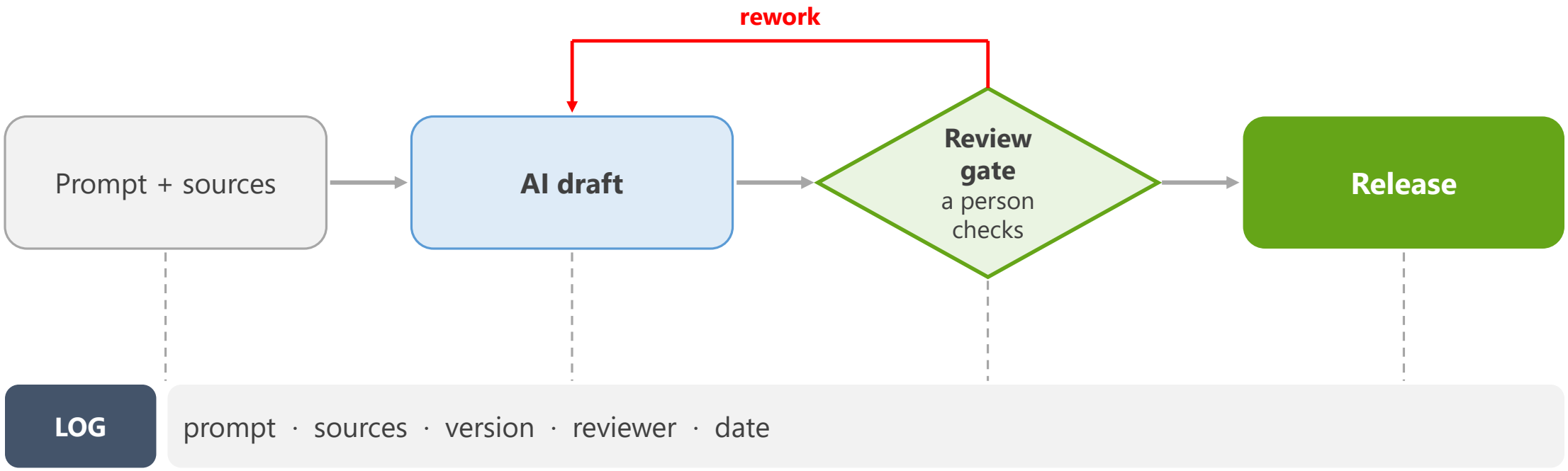


No label:
Who wrote it? Who checked it?

Labeled:
clear origin, owner and review

AI Governance
Audit Trail and Review Gate

- Nothing AI-made leaves the team without a human review
- Keep a light log so decisions stay traceable



AI Governance
Bias in AI Outputs

- AI repeats patterns from its training data and our prompts
- Check results before they affect people or decisions



Techniques
Ask before Answering

- Guide the LLM to ask for clarification before giving an answer.
- Helps ensuring that the model's answers are as accurate and specific as possible.

- **Prompt Formula**
    1. You are an expert in the field of *[industry]*. I'm going to ask you to complete some specific tasks, but before you answer, I want you to do the following: If you have any questions about my task or uncertainty about delivering the best answer possible, always ask bullet point questions for clarification before generating your answer. Is that understood?
    2. Great, my question is *[question]*. Your task is to *[task]*. Please ask any questions you have so that I can improve my prompt before you complete your task.

Techniques
Using Different Perspectives

- Technique that prompts the LLM to generate responses from diverse perspectives (e.g., user, expert, critic) before synthesizing them into a balanced, comprehensive answer.
- **Why it matters**:
  - Single-perspective answers can miss critical nuances, leading to biased or incomplete solutions.
  - Framing your prompt from a specific viewpoint forces the LLM to pull specific terminology, constraints, and priorities associated with that role.

- **Define Stakeholders**: Identify relevant perspectives (e.g., user, expert, critic).
- **Generate Perspectives**: Produce individual responses from each viewpoint.
- **Synthesize**: Combine insights into a balanced, cohesive final answer.
- **Evaluate Balance**: Ensure the final output addresses key concerns from all perspectives.

CHALLENGE 4

The AI Working Agreement

"Kim needs a one-page AI working agreement by Friday. The PO, the dev lead and the works council already disagree."

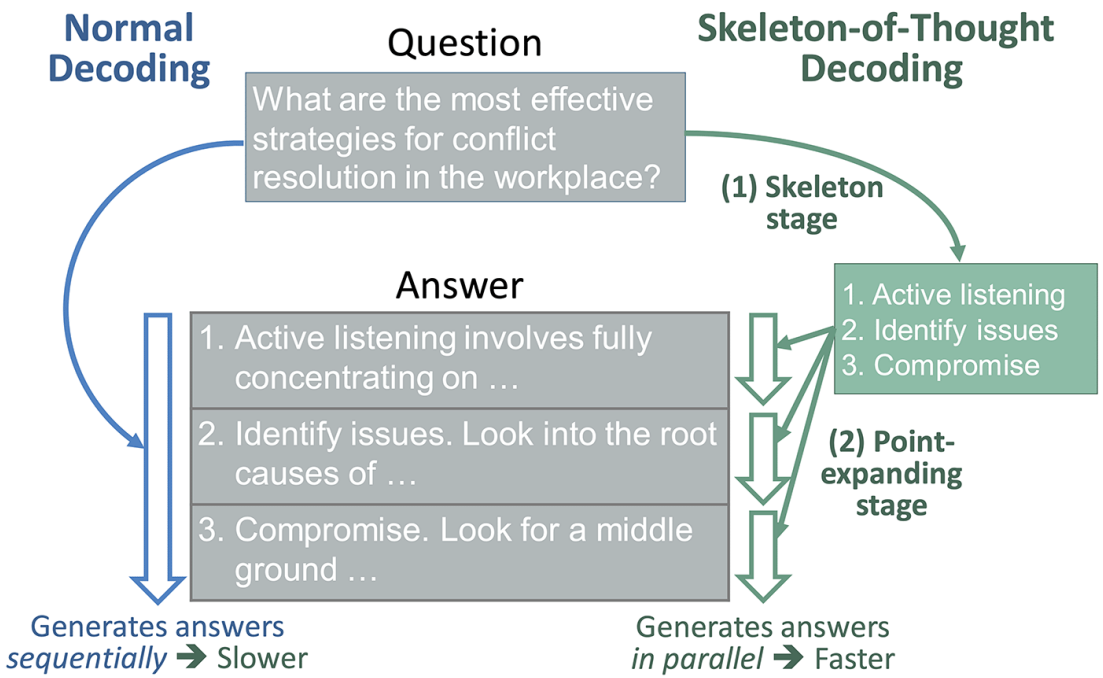
Clean a data export first, then draft the agreement and save it as a Kiro steering file.

Audience, Depth and Multi-Input Prompts

Techniques
Skeleton-of-Thought

- Designed to reduce AI response latency and improve structural quality.
- It mimics how **humans write** by **first** creating an **outline (the skeleton**) and then filling it.
- 2 Steps:

  (1) first asks the LLM to derive a skeleton of the answer.
  (2) asks the LLM to provide the answer to each point in the skeleton.

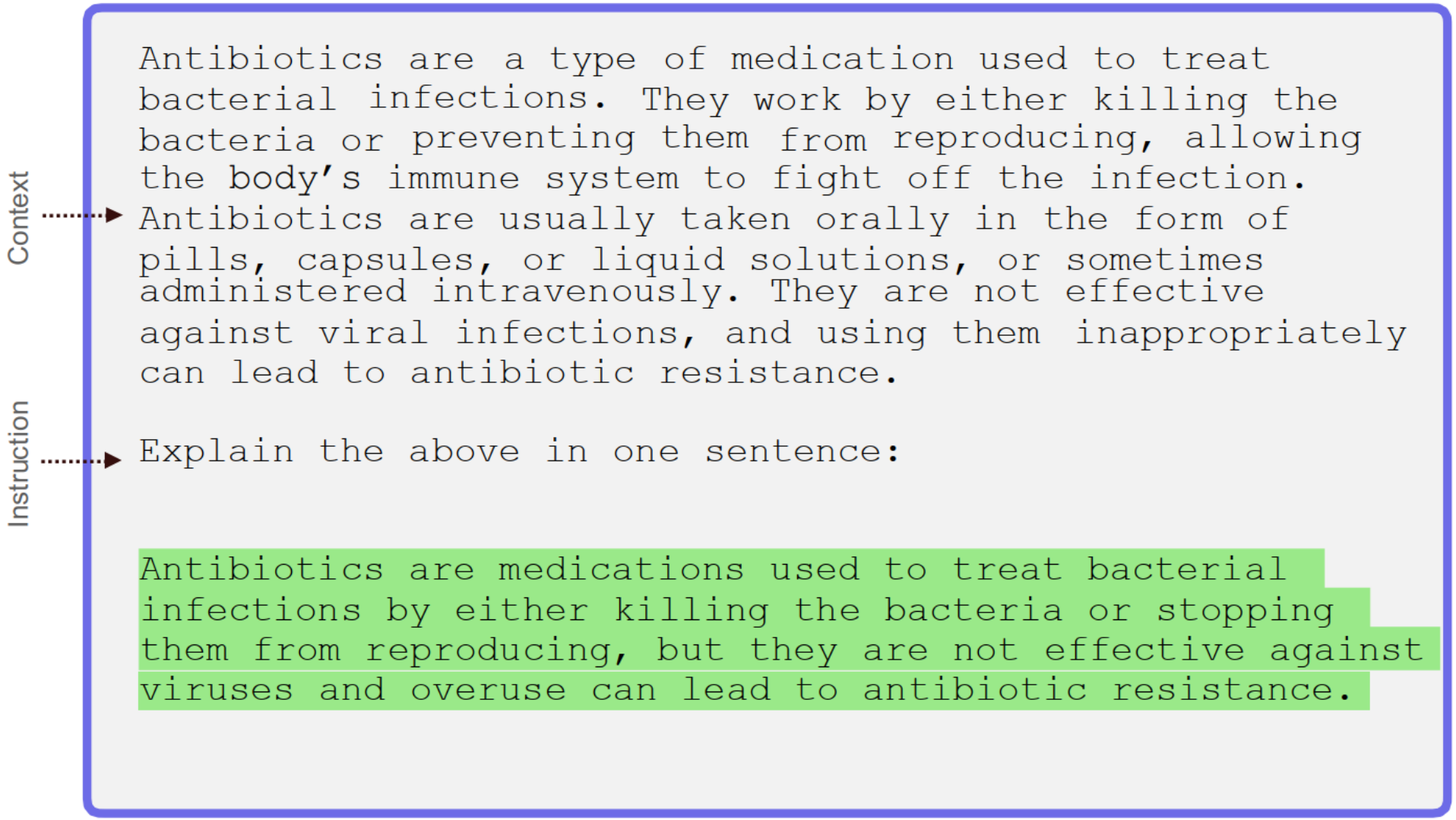


Techniques
Exclude Unwanted Outputs

- Direct the LLM to exclude certain parts of the text or specify what should be avoided.
- You can also exclude “chapters” or “sections” of documents.

- Blacklisting words:
  - Exclude - Write a summary about banks but **avoid using the word** loans
- Topic Constraints:
  - Exclude - Write a review on iphone **without covering** the price aspect
- Output type constraints:
  - Exclude - Write a poem about nature but **avoid using** rhyming words

Basics
Text Summarization



CHALLENGE 5

One Sprint, Three Very Different Readers

"*The same Sprint Review facts must become three updates: for the team, the steering committee and Rail Ops.*"

Draft all three with AI, then run them through your own review gate.

Application: Interface Specifications

Techniques
Activate Creativity

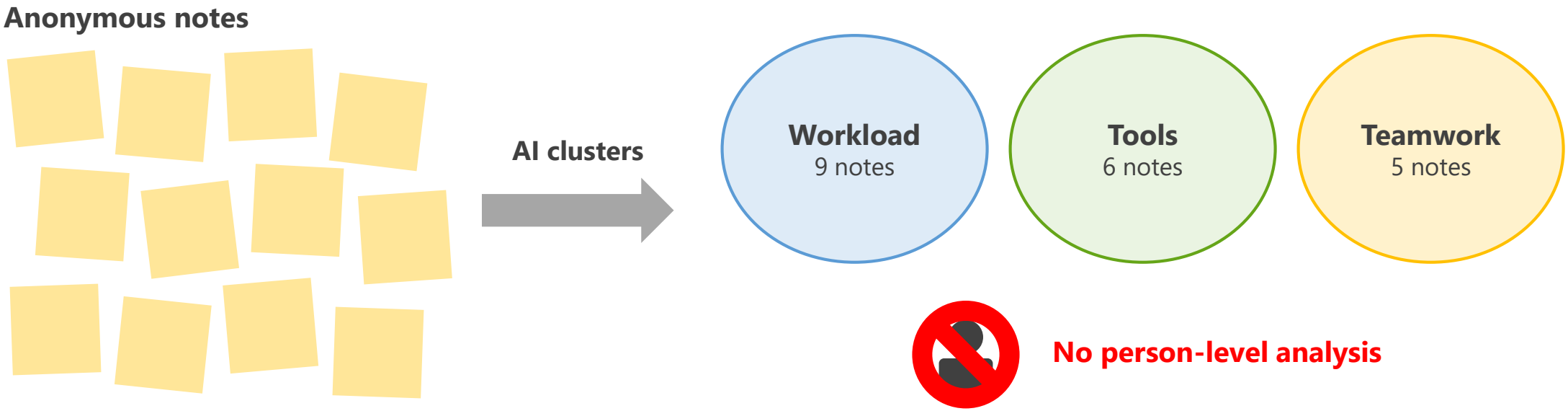
- Guide the LLM to bring in creativity.
- Can help you generate new ideas.

- **Prompt Formula**
  - *[Your content]*. Do you have more ideas to this topic?
  - *[Your content]*. What do you think about it?
  - *[Your content]*. Did I miss important aspects?

- Example
  - I want to invest in stocks from VW. What do you think about that?
  - I have the following application *[text]* to the job description *[description]*. Did I miss important aspects?

Techniques
Clustering and Classifying Text

- **Classify**: you set the categories, AI sorts the notes
- **Cluster**: AI proposes themes, you check and name them
- **Team level only**: never analyze who wrote what



CHALLENGE 6

The Retro Nobody Wants Again

"*The team is bored of the same retro, forty anonymous sticky notes are waiting, and a manager wants to know who is the most negative.*"

Design a fresh retro, cluster the notes by topic and mood, and derive three actions.

Working with Documents and Constraints

Techniques
Delimited Prompting

What are Delimiters?

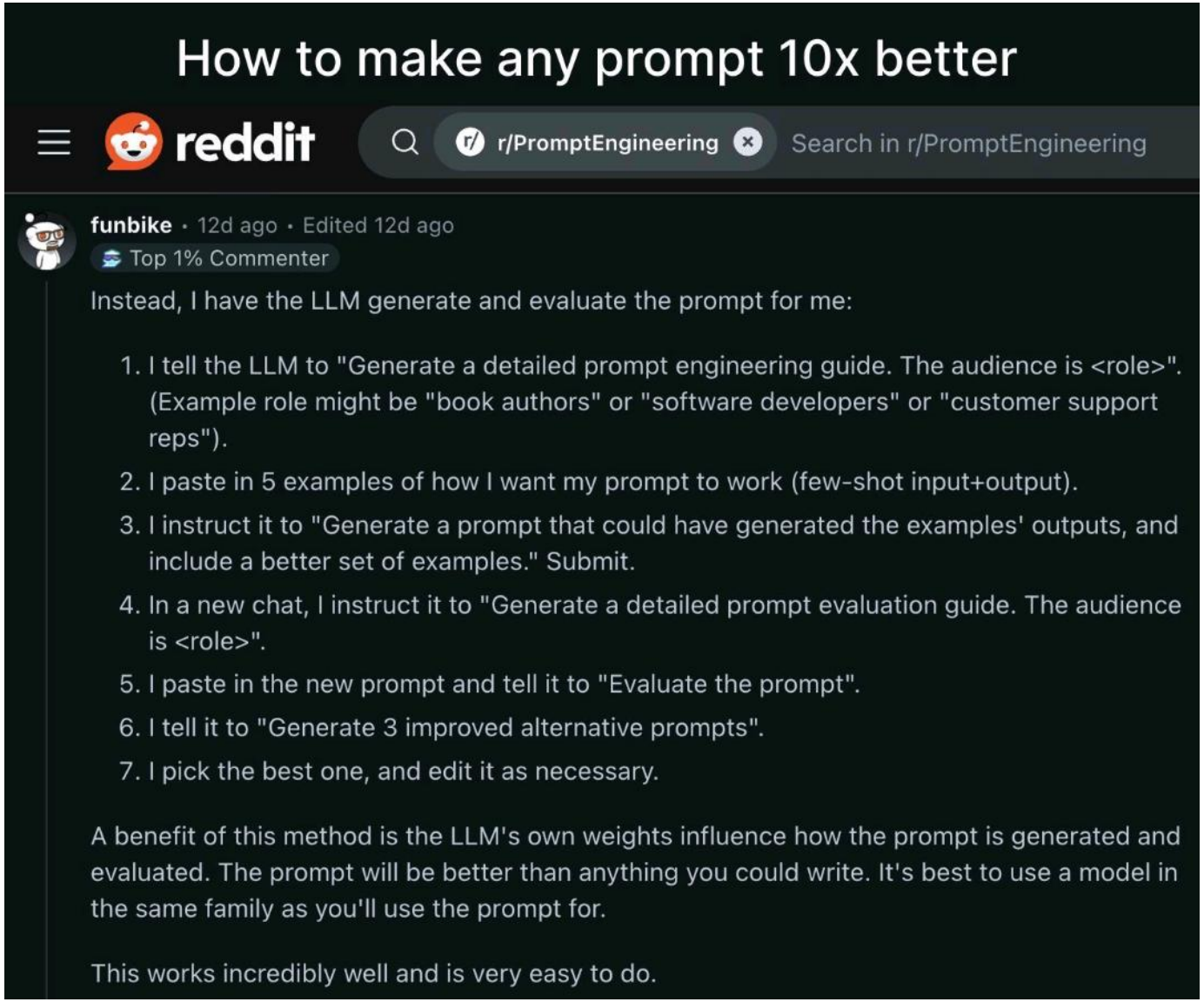
In prompt engineering, delimiters are specific characters or sequences used to separate different parts of a prompt. These can be simple like triple quotes (""""), XML tags (<tag> </tag>), or unique symbol combinations (###, ===, >>>). Delimiters act as markers, enabling the LLM to identify and interpret distinct sections within the prompt, such as instructions, context, input data, or desired output format.

Techniques
Delimited Prompting

- **Clarity:** They enhance the clarity and organization of prompts, making it easier for both the LLM and the human user to understand the structure and intent.
- **Control:** By clearly separating different sections, delimiters provide greater control over how the LLM processes the input and generates the output.
- **Error Reduction:** Well-defined sections reduce ambiguity and the likelihood of the LLM misinterpreting the prompt.
- **Flexibility:** Delimiters allow for flexibility in structuring prompts, accommodating various levels of complexity and different types of tasks.
- **Structured Outputs:** Delimiters enable LLMs to generate structured outputs like lists, tables, and code blocks. By clearly defining the desired output format within the prompt, delimiters guide the LLM to produce suitable responses.

For Reuse: offload a prompt response template structure into a separate file and refer to it in the prompt!

Some Creative Ideas



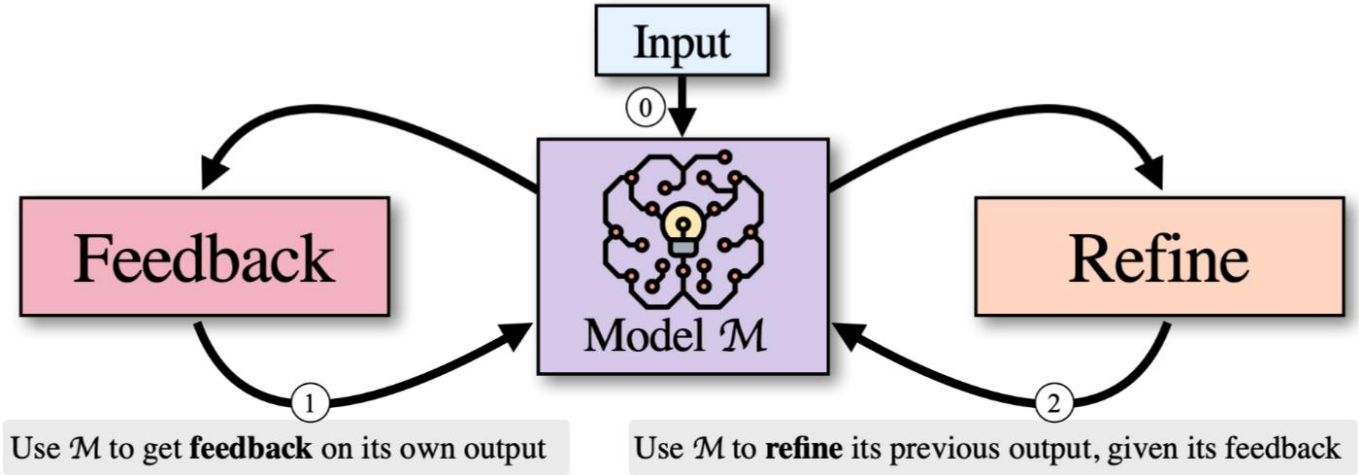
Techniques
Constructive Critic

- Provide objective and expert feedback on your writing.
- Highlighting areas for improvement and offering constructive criticism to help you refine and enhance your work.

- **Prompt Formula**
  - I want you to act as an expert and critic in the subject of *[industry]*. Criticize my content pasted below, convince me why it's bad, and give me constructive criticism on how it should be improved. For some context, my *[product or service]* is for *[details, demographic, etc]*. The purpose of my *[product or service]* is to *[your content goal]*. Let's think step by step and I want you to address each piece of content individually. Here is my content to critique, *[your content]*.

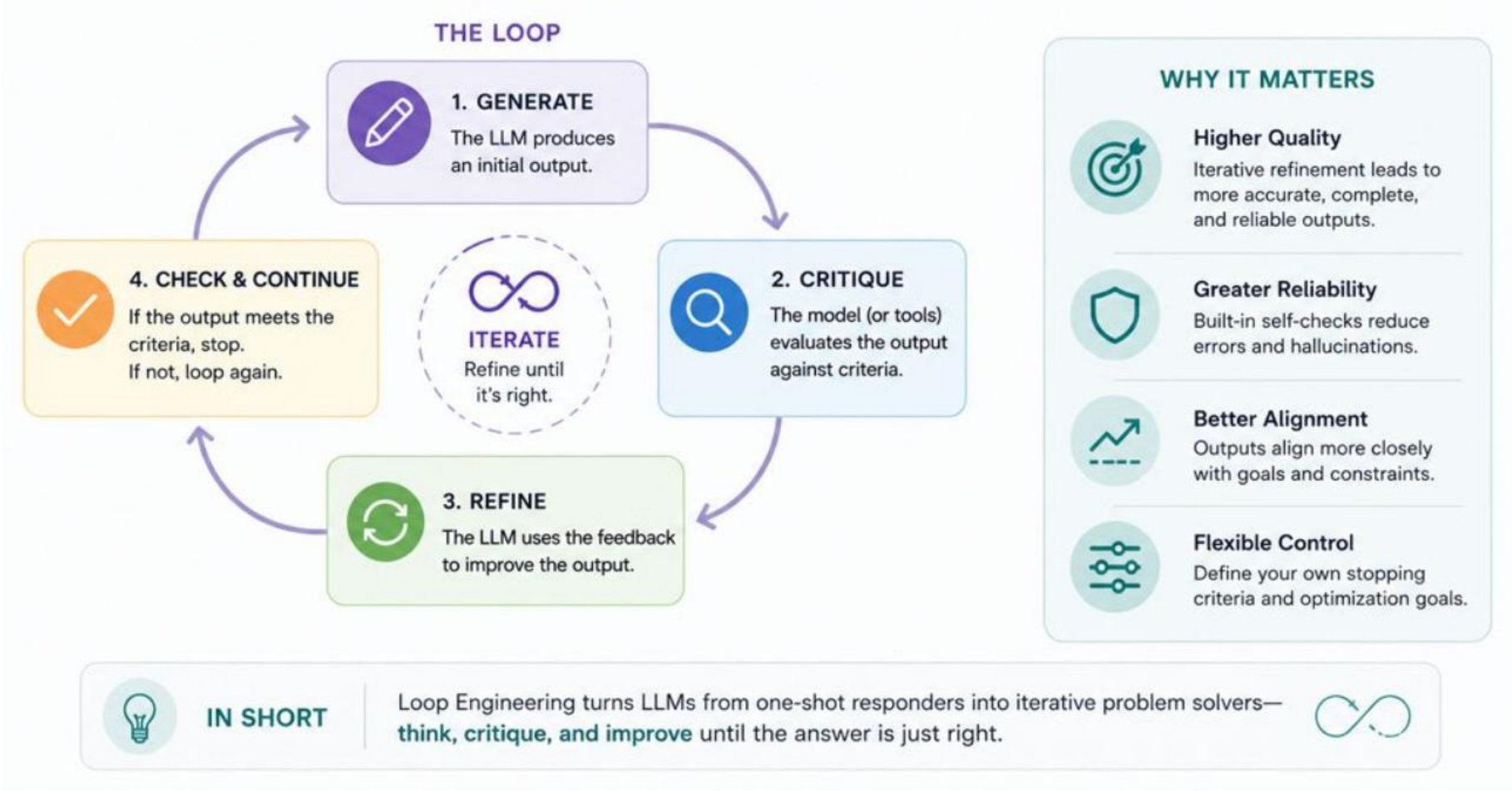
Techniques
Self-Refine

- Inspired by humans' ability to refine the solution, Self-Refine aims to improve the initial outputs from LLMs.
- Process:
  1. **Initial output**: Prompt the model to get the initial output.
  2. **Feedback**: Pass the prompt and initial output back to the model to get the feedback.
  3. **Refinement**: Pass the feedback back to the model to get the refined output.



Techniques
Loop Engineering and Iterative Refinement

- Use the LLMs or agents chat context to continuously iterate and improve the prompts.



CHALLENGE 7

Write It Down Before It's 'How We've Always Done It'

"Kim can't find last week's decisions in a messy wiki, and one decision from week one was never documented."

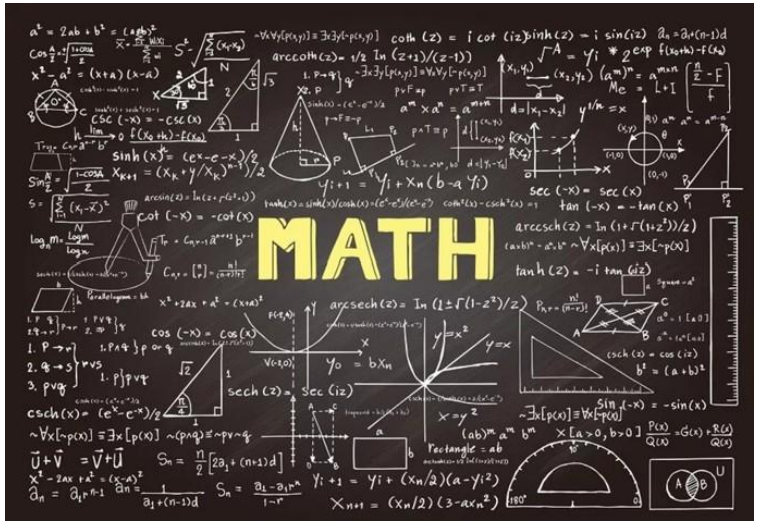
Build a template and knowledge page with AI, then write the decision record.

Structured Reasoning and Validation

Math in LLMs

- An LLM is not really calculating results as we humans do.
- It performs pattern matching and tries to transfer the relationship between numbers to the current task (Reasoning).

- **Ensure you double check any task where the LLM has to calculate something!**



Techniques
AI with Data: Formulas, Not Mental Math

- AI writes the formulas and explains each step
- Excel does the calculation, you check the result

1 AI writes the formula

Write the Excel formula for weekly throughput. Don't calculate the result yourself.

```
=COUNTIFS(D:D,">="&A2,D:D,"<"&A2+7)
```

2 Excel calculates

Week start	Done
01 Sep	6
08 Sep	4
15 Sep	7

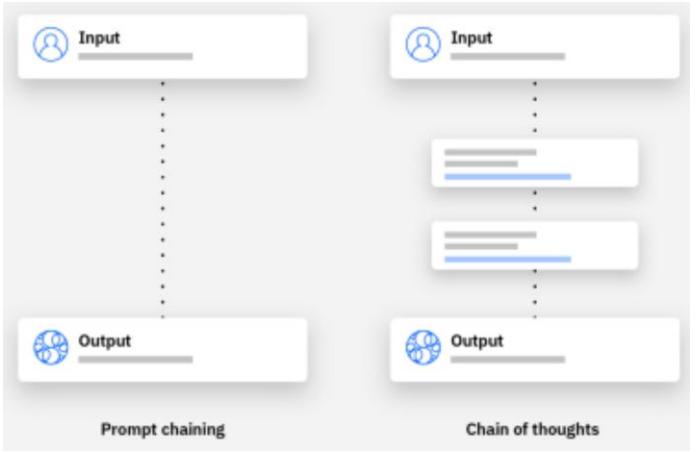
3 You check



Sanity check: does the total match the export?

Techniques
Chain-of-Thought (CoT) Prompting

- Technique to guide the LLM to explain its answers in a step-by-step manner.
- Helps to gain transparency of the answering process.



- Prompt Formula
  - *[Your question]*. Let's think step by step.
- Examples
  - What is the diameter of the sun? Let's think step by step.
  - What is the weight of an oxygen molecule? Let's think step by step.

Techniques
Zero-Shot Reasoning and CoT

Before

What is the annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day?

The annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day is 8,000,000 liters per year.

After

What is the annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day?

Let's think step by step and explain the calculation step by step.

Step 1: Calculate the total number of days in a year:
There are 365 days in a year.

Step 2: Calculate the total number of days the family is at home:
The family is at home an average of 200 days per year.

Step 3: Calculate the total water demand of the family:
The family uses an average of 100 liters of water per day. Therefore, the total water demand of the family is 100 liters x 200 days = 20,000 liters per year.

Step 4: Calculate the annual water demand of the single-family household:
The annual water demand of the single-family household is 20,000 liters per year.

Techniques
Plan-and-Solve Prompting

- **Chain-of-Thought (CoT) prompting**: blends planning and execution in a single stream of reasoning.
- **Plan-and-Solve**: explicitly decouples these stages.

Phase 1: Plan
Ask the model to break down the problem into a sequence of logical steps without performing any calculations.
"What steps are needed to solve this?"

Phase 2: Solve
Using the generated plan as a guide, execute each step in order to arrive at the final answer.
"Now, follow the plan and compute the result."

By isolating planning from execution, the model avoids getting distracted by intermediate computations and maintains focus on the overall strategy.

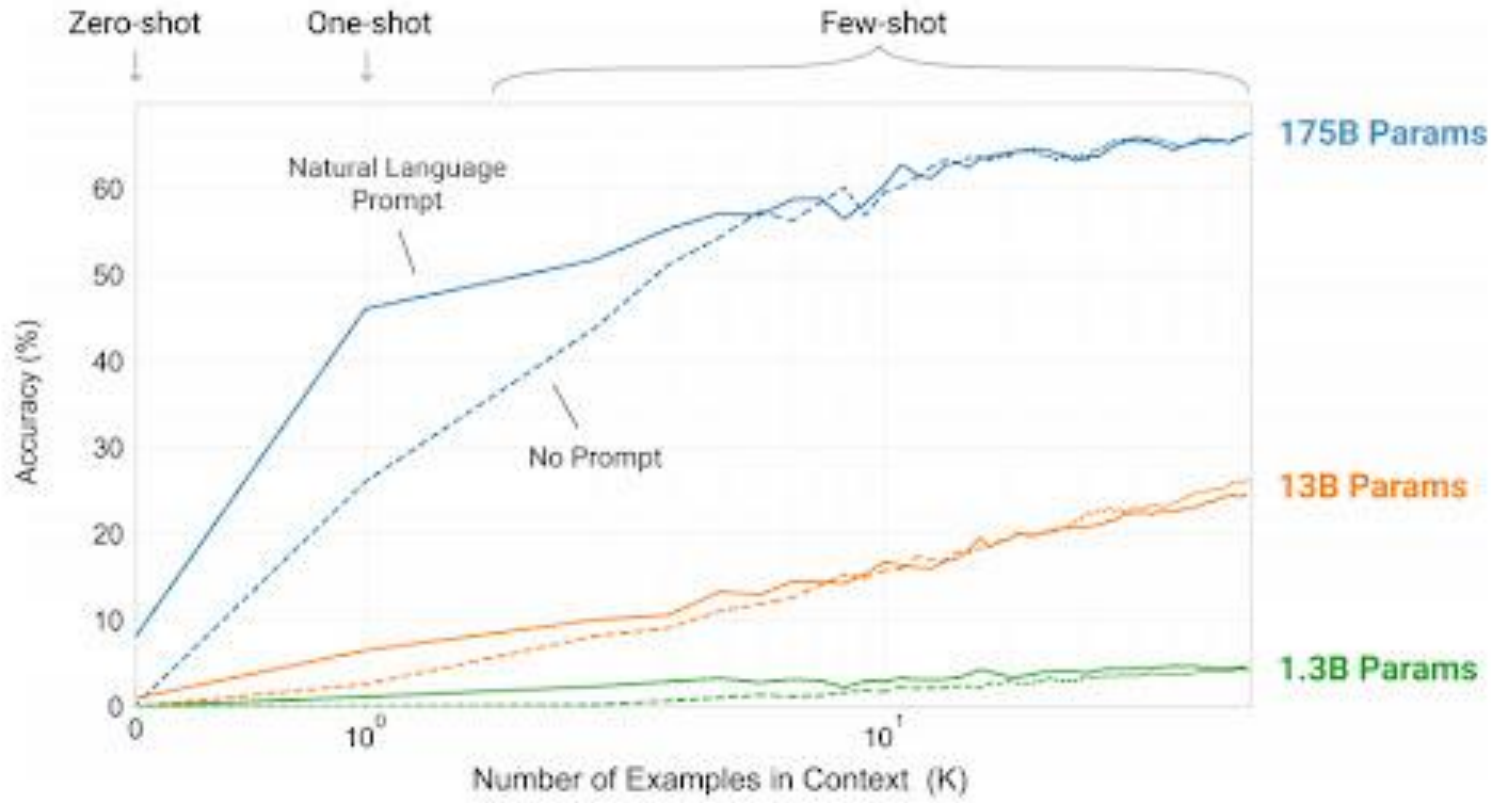
CHALLENGE 8

The Number Sascha Will Actually Believe

"Finance wants one defensible forecast. The Jira export has gaps, and two estimates disagree: four weeks or ten?"

Check the data, let AI write the formulas, let Excel calculate, then back-test.

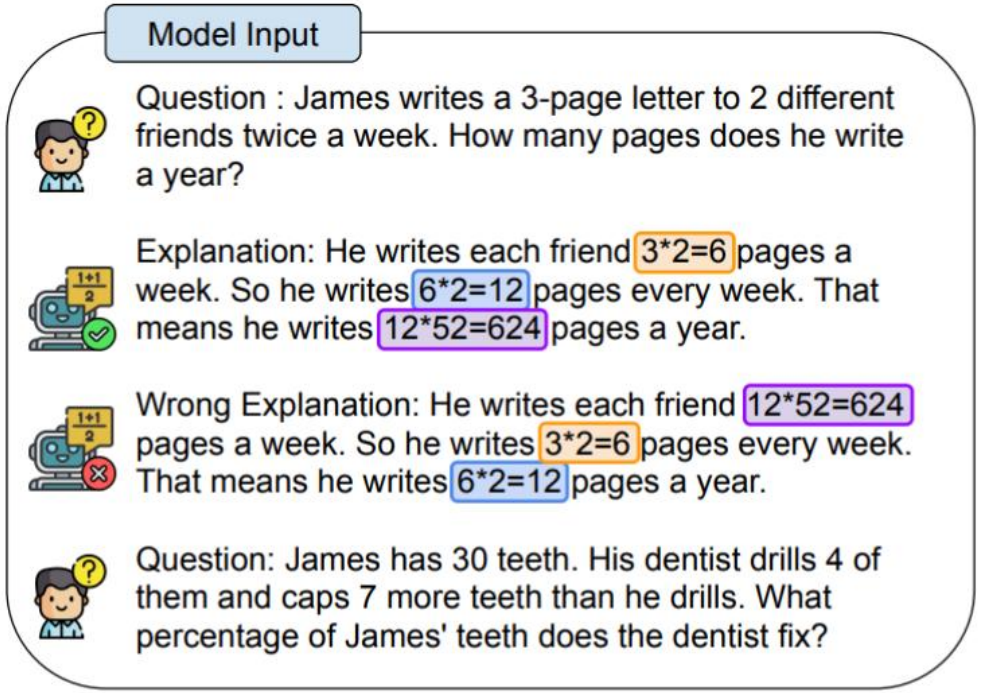
Techniques
Recap: Context and Accuracy



- As more context you provide in the input, as better the transformer can derive the context.
- As more „shots"/prompts you provide, as more accurate the result will be.

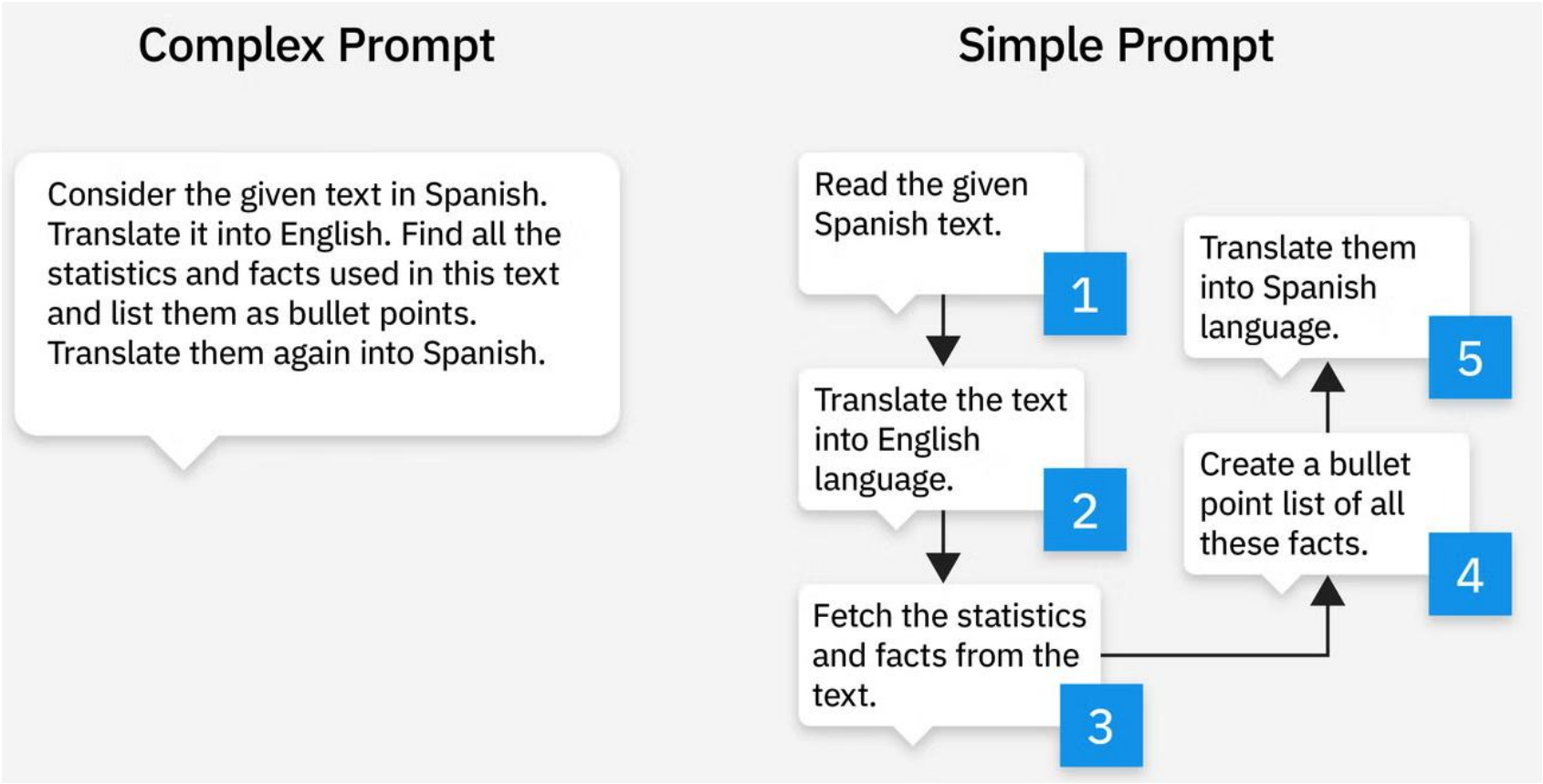
Techniques
Contrastive Prompting

- Chain-of-Thought Prompting improves accuracy in LLM reasoning.
- Instead of let the CoT reason by itself, we simply provide **positive and negative** examples as in Few-Shot prompting.
- Learning from failures is not onlv a human-like method to learn.



Techniques
Prompt Chaining

- Decompose complex tasks/prompts into several smaller steps and chain them.
- It prevents the model from losing focus or context on complex objectives.



CHALLENGE 9

Ask Jira, Then Automate With Brakes

"Kim wants five Jira answers and a weekly report by lunch, and the PO wants blocked stories escalated automatically."

Write and check the JQL, then design the automation with approval and log.

AI Governance
Legal Guardrails for People Data

- People topics are high-stakes: AI drafts, people decide
- Check DB rules and work agreements before you start

§ **GDPR**
No significant decisions made solely by automation (Art. 22)

§ **EU AI Act**
AI that evaluates or decides about staff can be high-risk

§ **Works council**
Tools that can monitor performance need its approval (§87 BetrVG)

OK with care
Team-level skill gaps · learning ideas · feedback drafts you finalize

Not OK
Ranking people · performance scores · automated decisions about people

CHALLENGE 10

Grow the Team, Not a Ranking

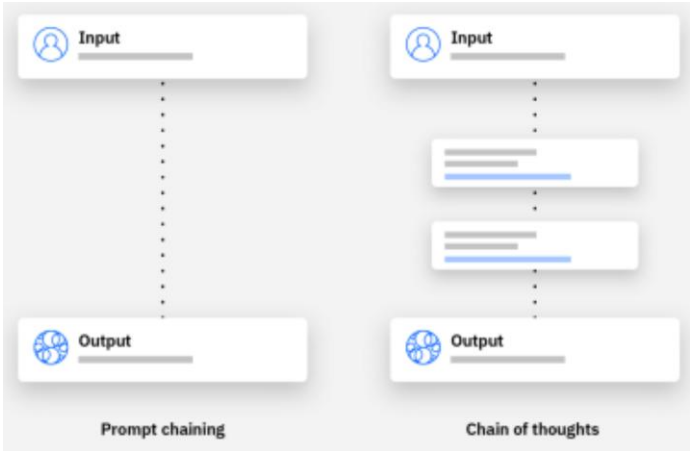
"The next PI needs skills the team doesn't fully have, and someone just asked AI to rank the team."

Map the skill gaps, draft learning paths and fair feedback, and decline the ranking.

Application: Architecture Decisions & ADRs

Techniques
Recap: Chain-of-Thought (CoT) Prompting

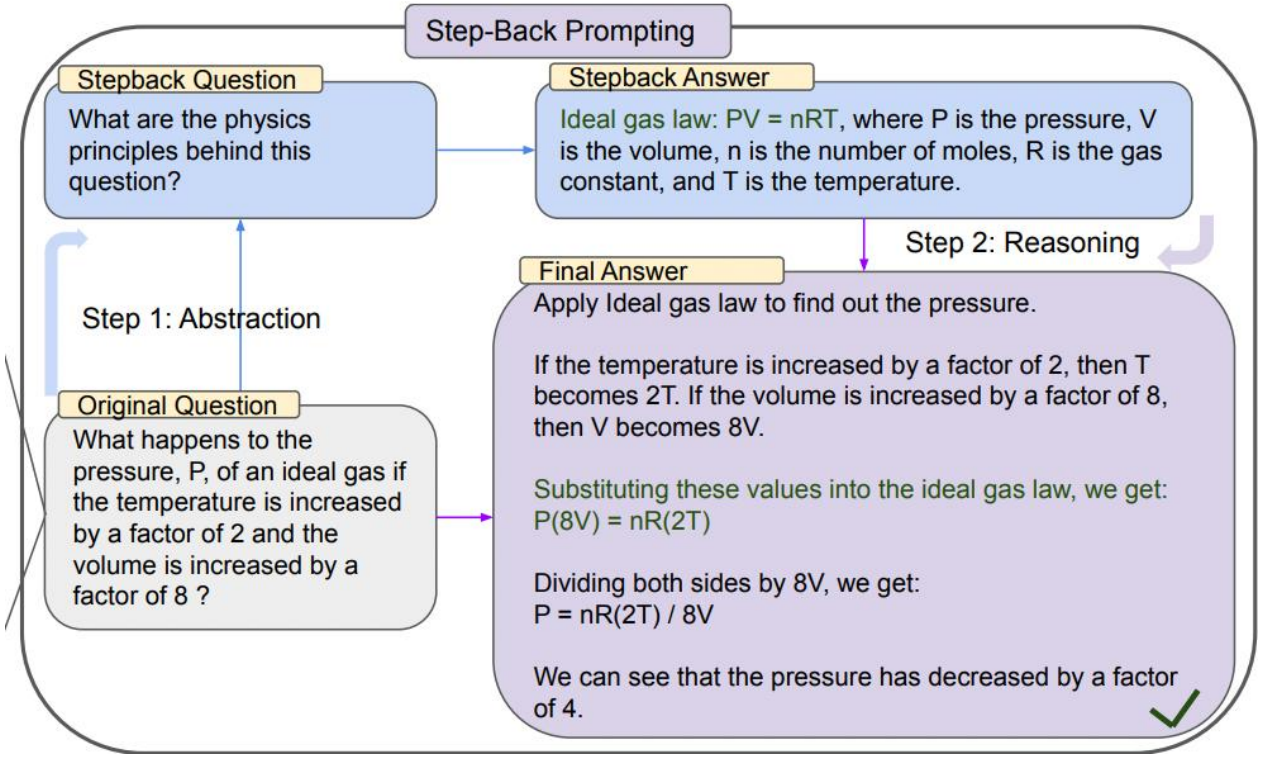
- Technique to guide the LLM to explain its answers in a step-by-step manner.
- Helps to gain transparency of the answering process.



- Prompt Formula
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- Examples
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  - What is the weight of an oxygen molecule? Let's think step by step.

Techniques
Step-Back Prompting

- Is a technique encouraging LLMs to "step back" and focus on abstract principles before reasoning through details.
- Prompting the model to first derive **high-level concepts** and **first principles** before tackling specific details.



CHALLENGE 11

Everyone Rated These Differently

"*Three business owners scored the same ten features differently, and then Team Signal loses two people.*"

Build one WSJF ranking with a decision check, then compare three PI scenarios.

CHALLENGE 12

Nothing Ships in Isolation

"All four teams bring their backlogs to the table. What depends on what, and who needs to talk to whom?"

Find the dependencies, score the risks and write an escalation that holds up.

CHALLENGE 13

The PI Readiness Pack

"Tomorrow Kim presents the train's readiness to the steering committee. One page, three slides and one sentence must tell the same story."

Reuse your results, pass your review gate and convince the jury.

CHALLENGE 14

Closing the Loop

"*Before you leave, you owe the two days an honest retro, including the parts that didn't go perfectly.*"

Run a short retro and pick three prompts or rules for your real team.